

Here Comes the Homology

Will we get there?

Linear Algebra Review

Field

A field $(k, +, \cdot)$ is a set k with two operations $+$ and $\cdot$, such that for any $a, b, c \in k$:

- Closure
- Commutativity
 -
 -
- Associativity
 -
 -
- Identity
 -
 -

- Inverses
 -
 -
- Distributivity

Examples:

$$(\mathbb{R}, +, \cdot)$$

$$(\mathbb{Z}, +, \cdot)$$

Vector Space

A vector space over a field k is a set V with vector addition and scalar multiplication such that

- Associative $+$:
- Commutative $+$:
- Identity:
 - $+$
 - $\cdot$
- Inverses $+$:

- Scalar vs. field mult:
- Distributivity
 -
 -

Examples:

$$V = \mathbb{R}^2$$

$$k = \mathbb{R}$$

Basis

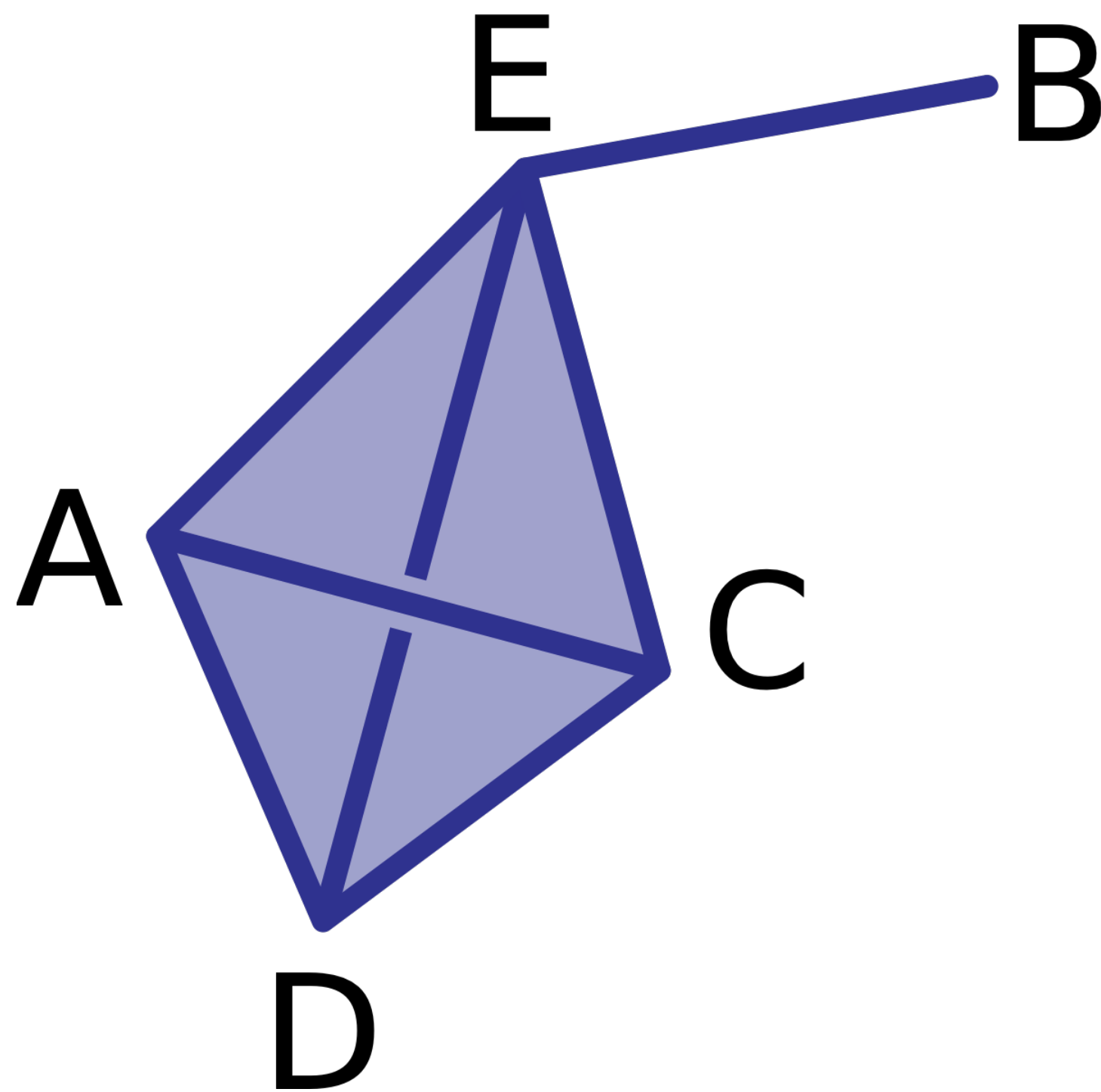
A basis for a vector space V is a collection of vectors $\{\mathbf{b}_\alpha\}_{\alpha \in A}$ such that

- They are linearly independent and
- They span V

p -Chains

Let K be a simplicial complex and fix a dimension p .

A p -chain is a finite formal sum of p -simplices in K , written $\alpha = \sum_i a_i \sigma_i$



Addition of Chains

p -chains are added component-wise: if $\alpha = \sum a_i \sigma_i$ and $\beta = \sum b_i \sigma_i$, then

$$\alpha + \beta =$$

Chain Group

The collection of p -chains with addition is called the p th-chain group (vector space), $C_p(K)$.

Some checks:

- Associative
- Commutative
- Zero element
- Inverses

Linear Transformation

A linear transformation between two vector spaces V and W is a map $T : V \rightarrow W$ such that the following hold:

- $T(v + w) = T(v) + T(w)$
- $T(a \cdot v) = a \cdot T(v)$

Matrix representation:

Boundary Maps

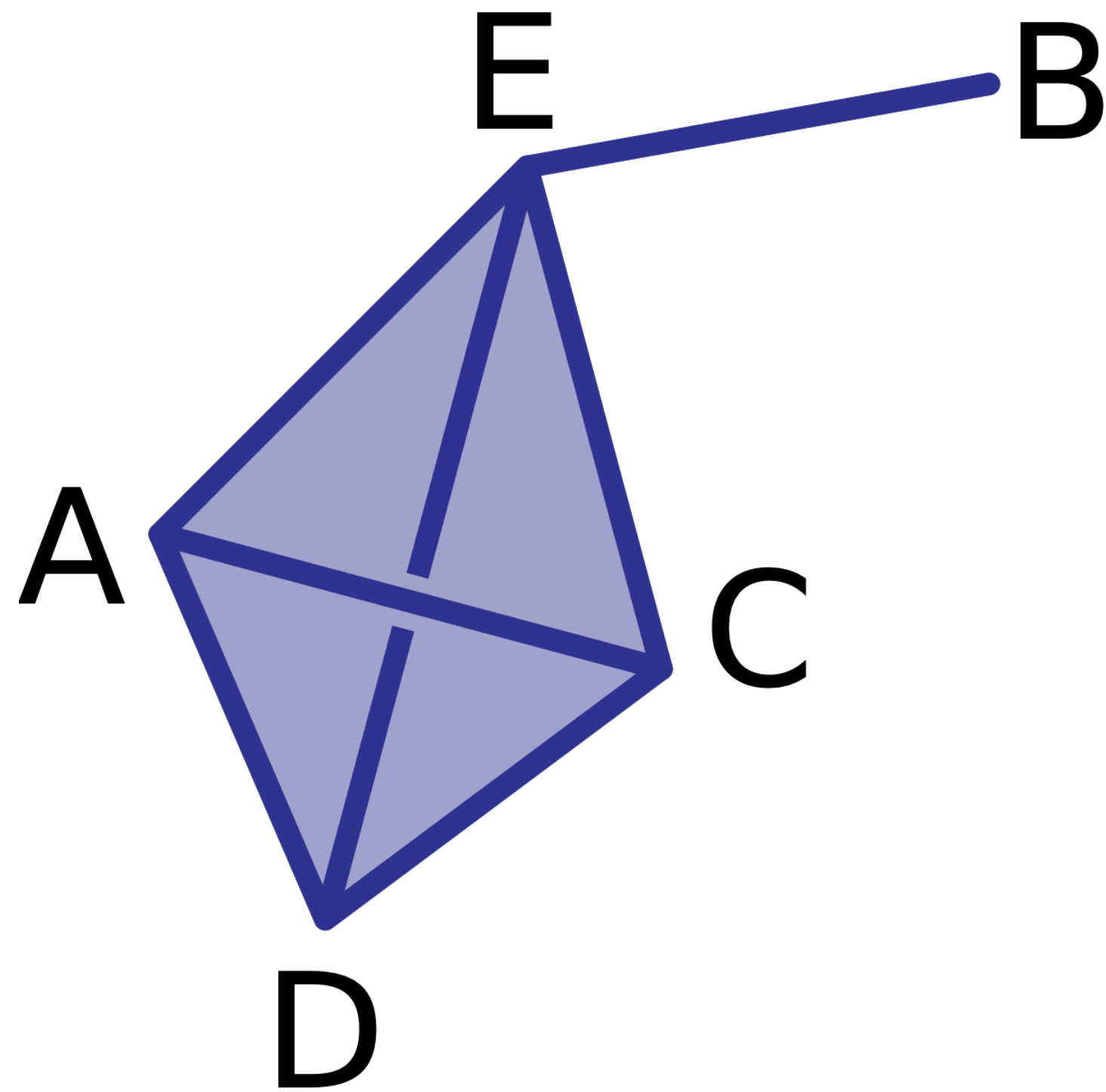
$$\partial_p : C_p(K) \longrightarrow C_{p-1}(K)$$
$$\sigma = [v_0, \dots, v_p] \longmapsto \sum_{j=0}^p [v_0, \dots, \widehat{v_j}, \dots, v_p]$$

We assume $\mathbb{Z}_2$ coefficients from now on!

Checking Linearity

$$\partial_p(\alpha + \beta) = \partial_p(\alpha) + \partial_p(\beta); \quad \alpha = \sum a_i \sigma_i, \quad \beta = \sum b_i, \sigma_i$$

Example



- $\partial_1([a, e]) =$

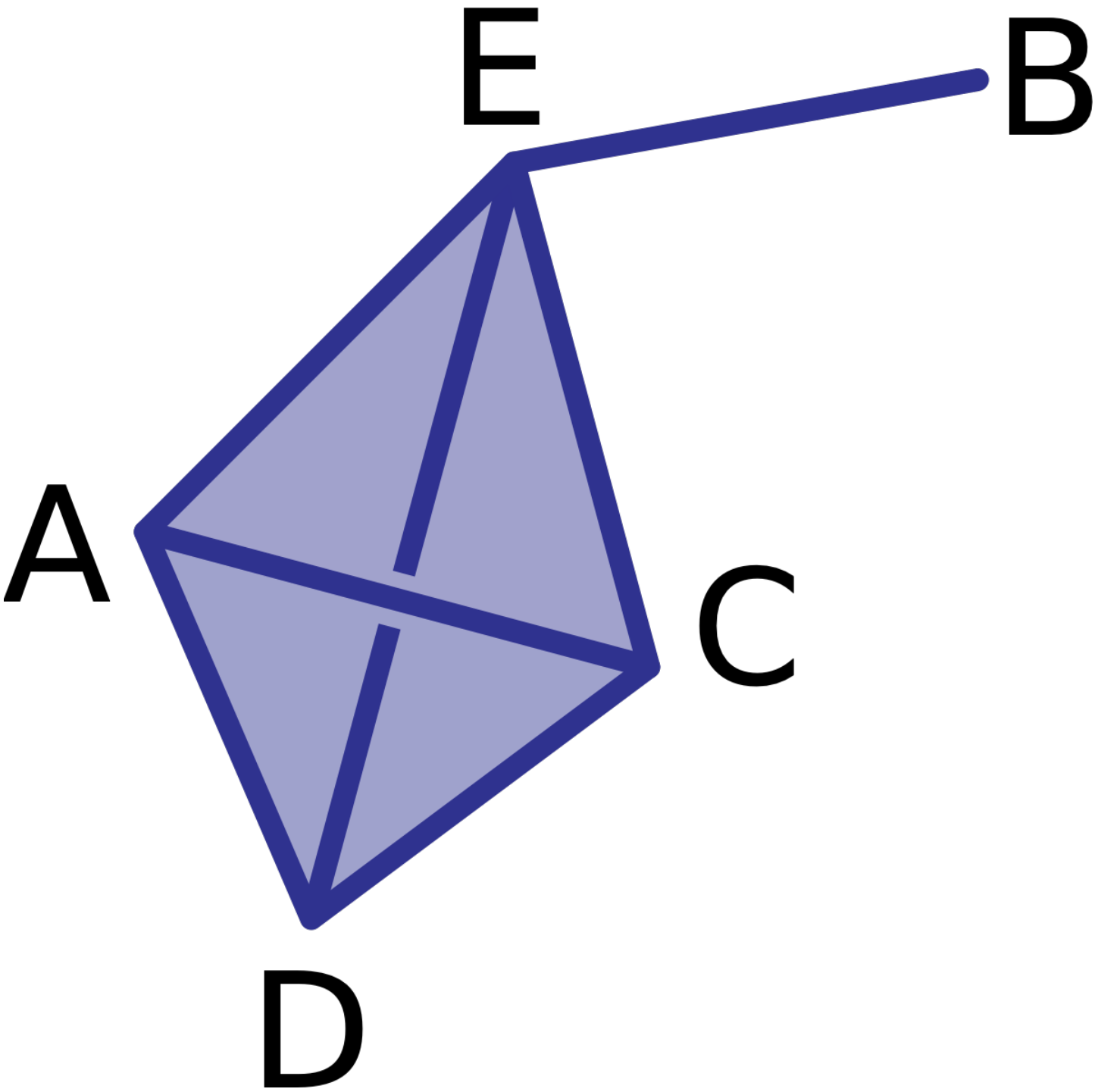
- $\partial_1([a, e] + [b, e]) =$

- $\partial_1([a, e] + [c, e] + [c, d] + [a, d]) =$

- $\partial_2([a, c, e] + [a, c, d]) =$

Matrix Representation

$$\partial_1(K) : C_1(K) \rightarrow C_0(K)$$



$\partial_1(K) =$

	AC	AD	AE	BE	CD	CE	DE
A							
B							
C							
D							
E							

Chain Complex

$$\cdots \xrightarrow{\partial_{p+2}} C_{p+1}(X) \xrightarrow{\partial_{p+1}} C_p(X) \xrightarrow{\partial_p} C_{p-1}(X) \xrightarrow{\partial_{p-1}} \cdots \longrightarrow C_0(X)$$

$$\partial_p : C_p(K) \longrightarrow C_{p-1}(K)$$

$$\sigma = [v_0, \dots, v_p] \longmapsto \sum_{j=0}^p [v_0, \dots, \widehat{v_j}, \dots, v_p]$$

Cycles and Boundaries

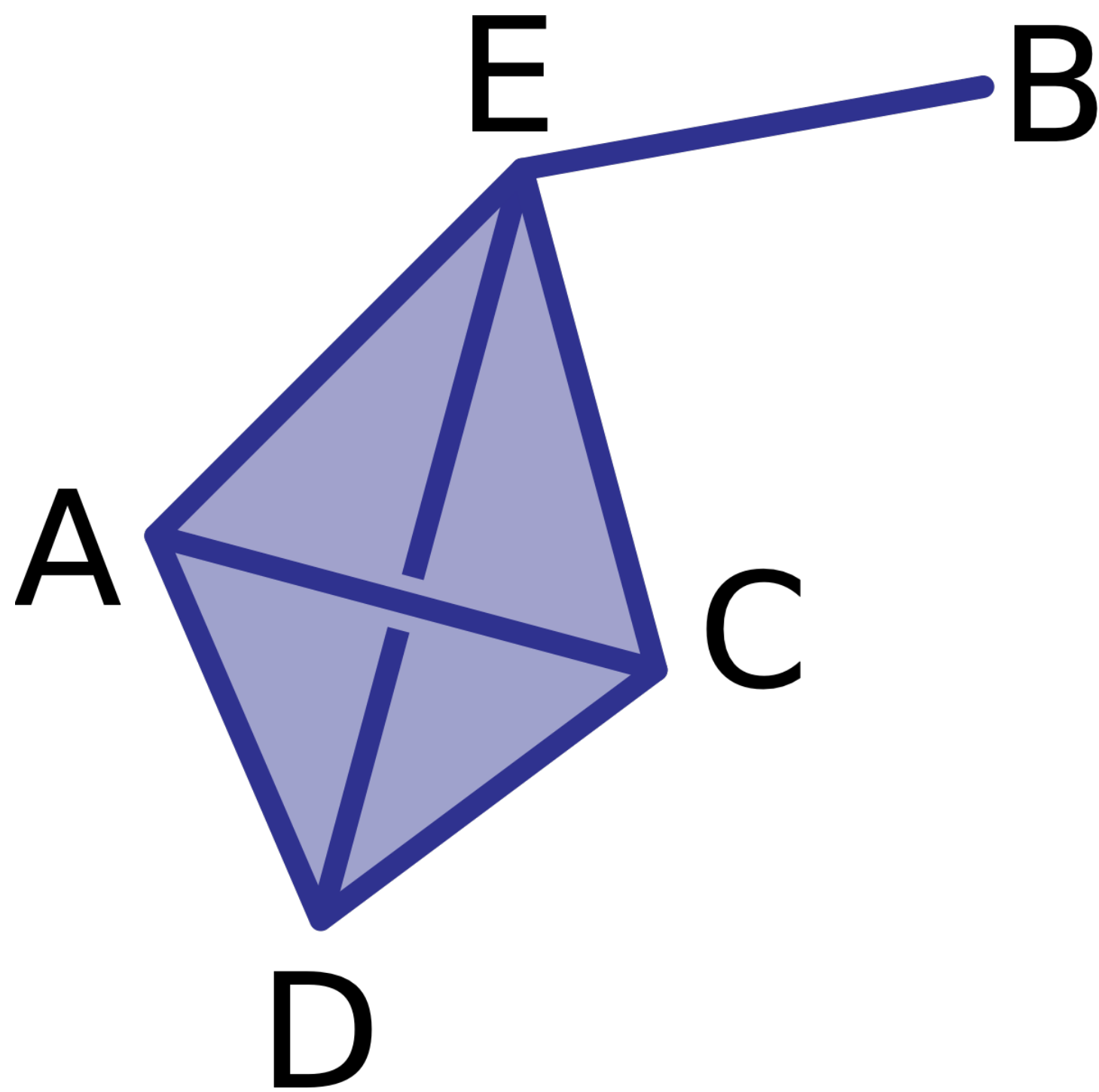
Subspaces for a Linear Transformation

$$T : V \rightarrow W$$

- Image:
- Kernel

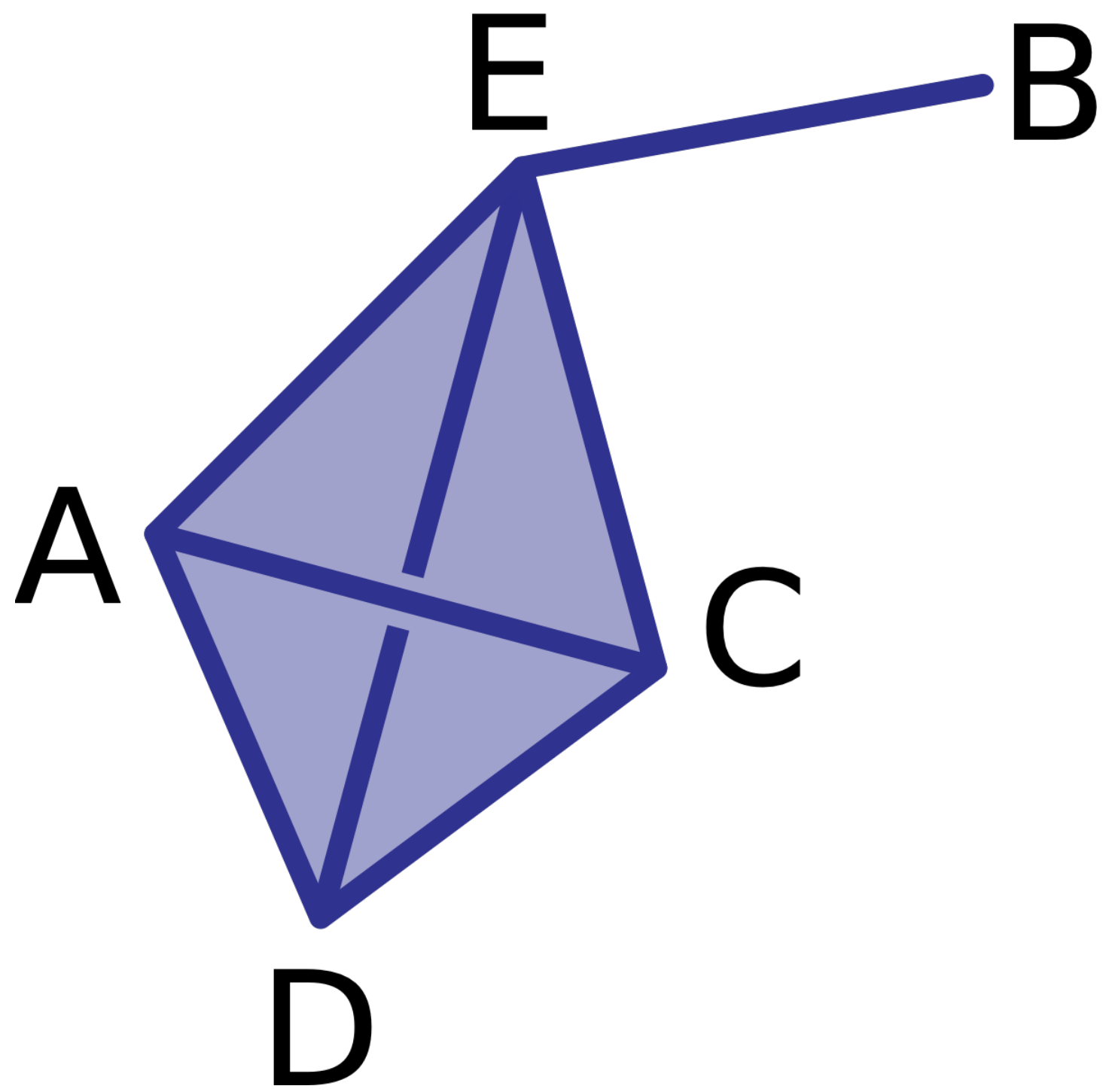
Cycles

A chain in the kernel of ∂_p is called a p -cycle. $C_{p+1}(K) \xrightarrow{\partial_{p+1}} C_p(K) \xrightarrow{\partial_p} C_{p-1}(K)$



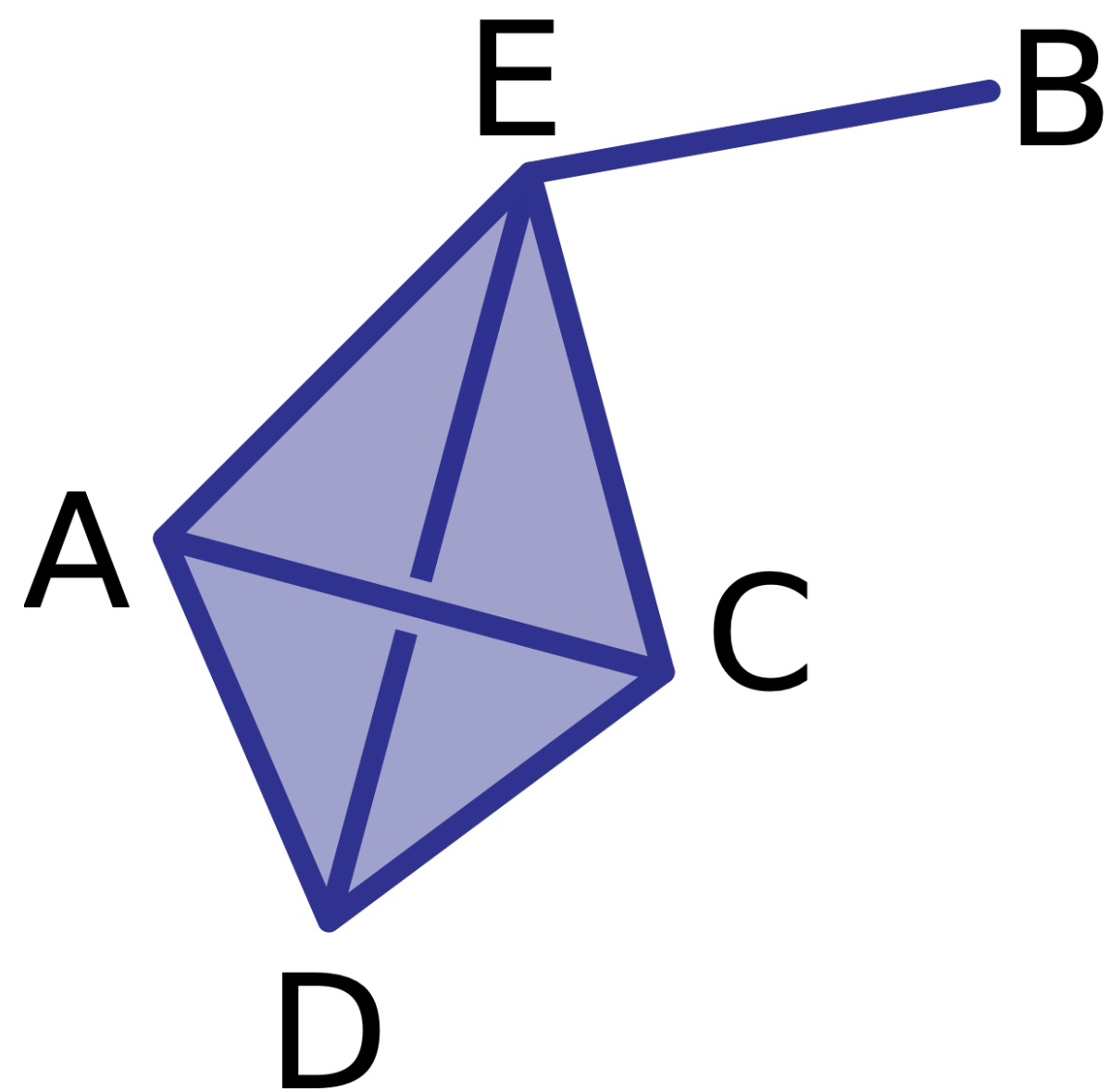
The collection of p -cycles forms a subspace $Z_p(K) \subseteq C_p(K)$

What is a 2-Cycle?



Boundaries

A chain in the image of ∂_{p+1} is called a p -boundary



$$C_{p+1}(K) \xrightarrow{\partial_{p+1}} C_p(K) \xrightarrow{\partial_p} C_{p-1}(K)$$

The collection of p -boundaries forms a subspace $B_p(K) \subseteq C_p(K)$

Nifty Trick

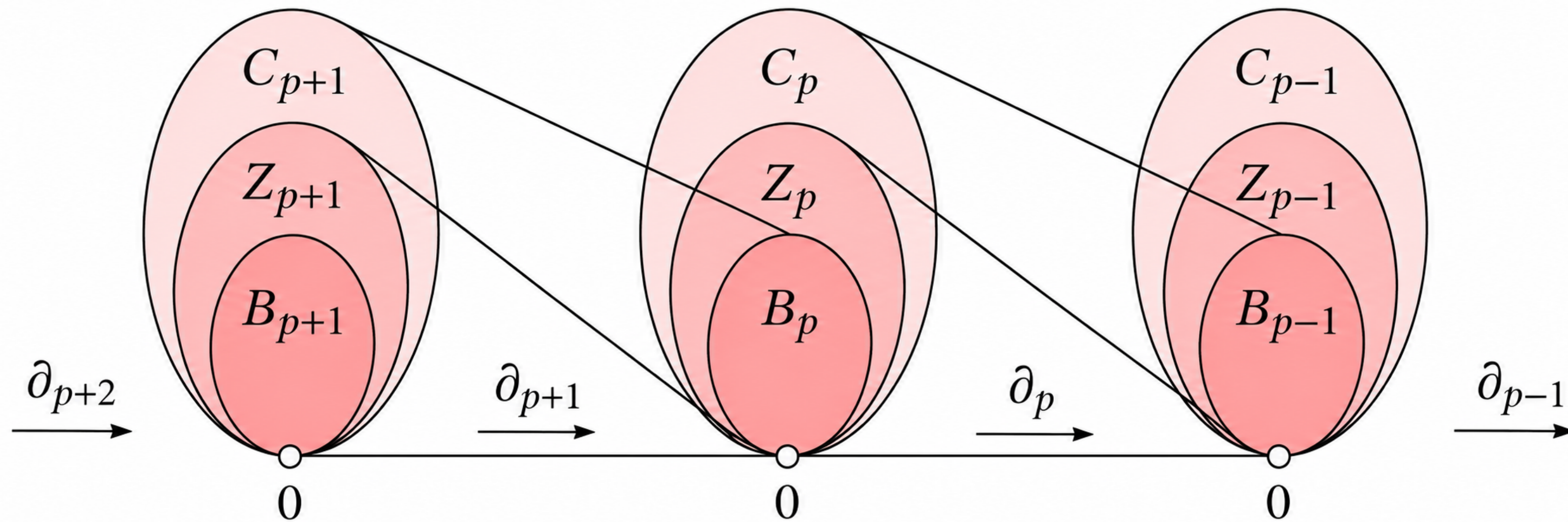
Theorem: $\partial_p \partial_{p+1}(\alpha) = 0$ for every $(p+1)$ -chain α

Translation

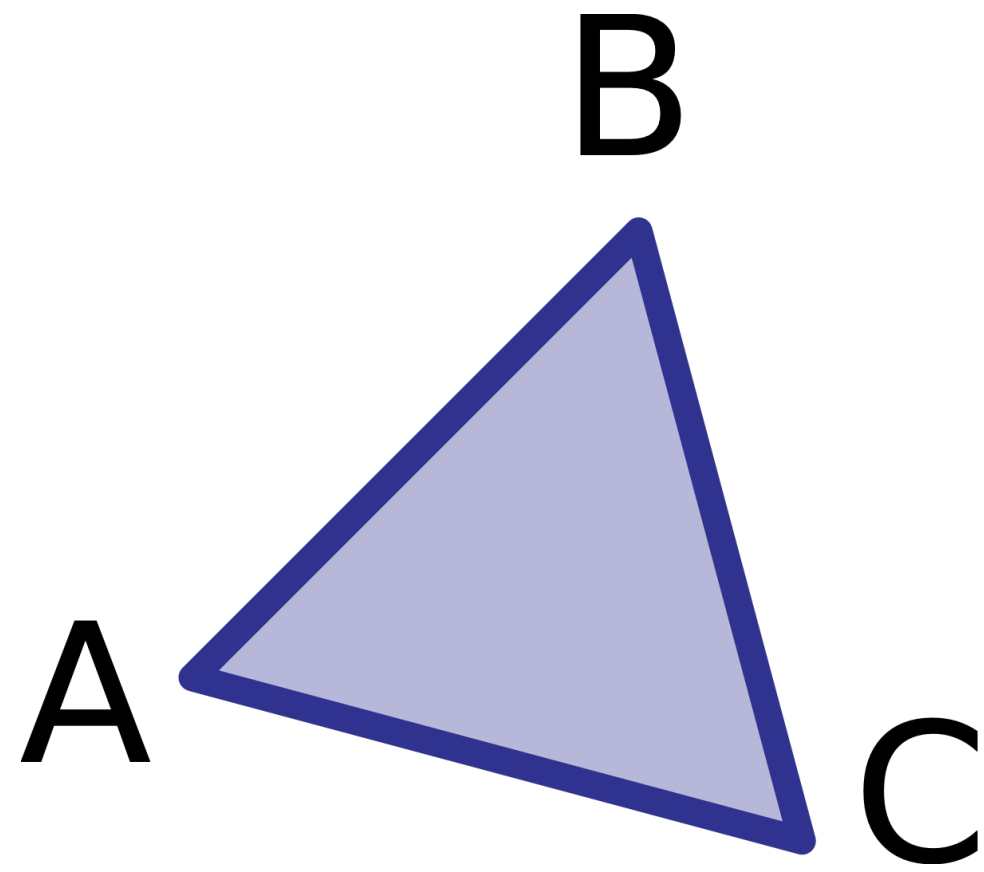
Every p -boundary is a p -cycle.

$$B_p(K) \subseteq Z_p(K) \subseteq C_p(K)$$

$$C_{p+1}(K) \xrightarrow{\partial_{p+1}} C_p(K) \xrightarrow{\partial_p} C_{p-1}(K)$$

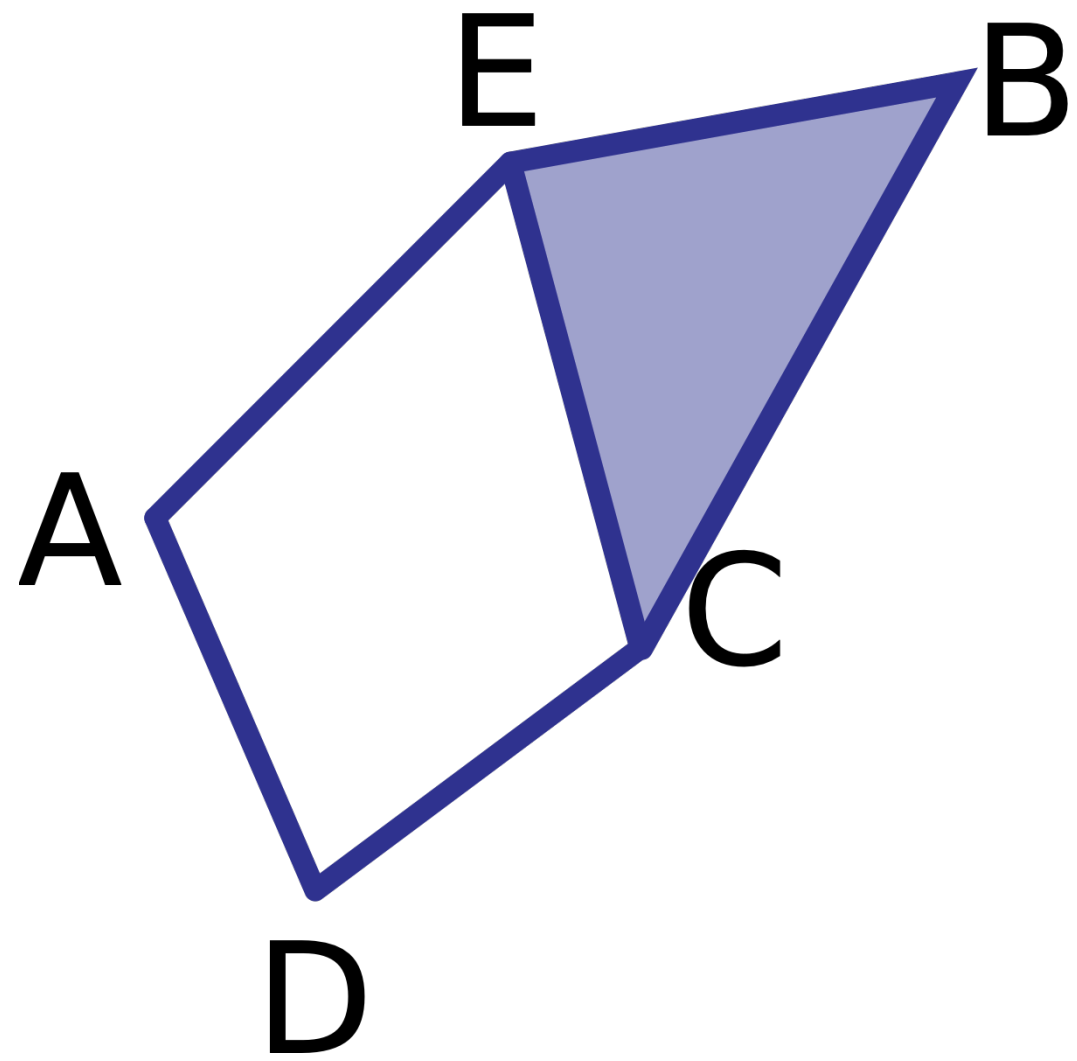


Example: Cycles and Boundaries



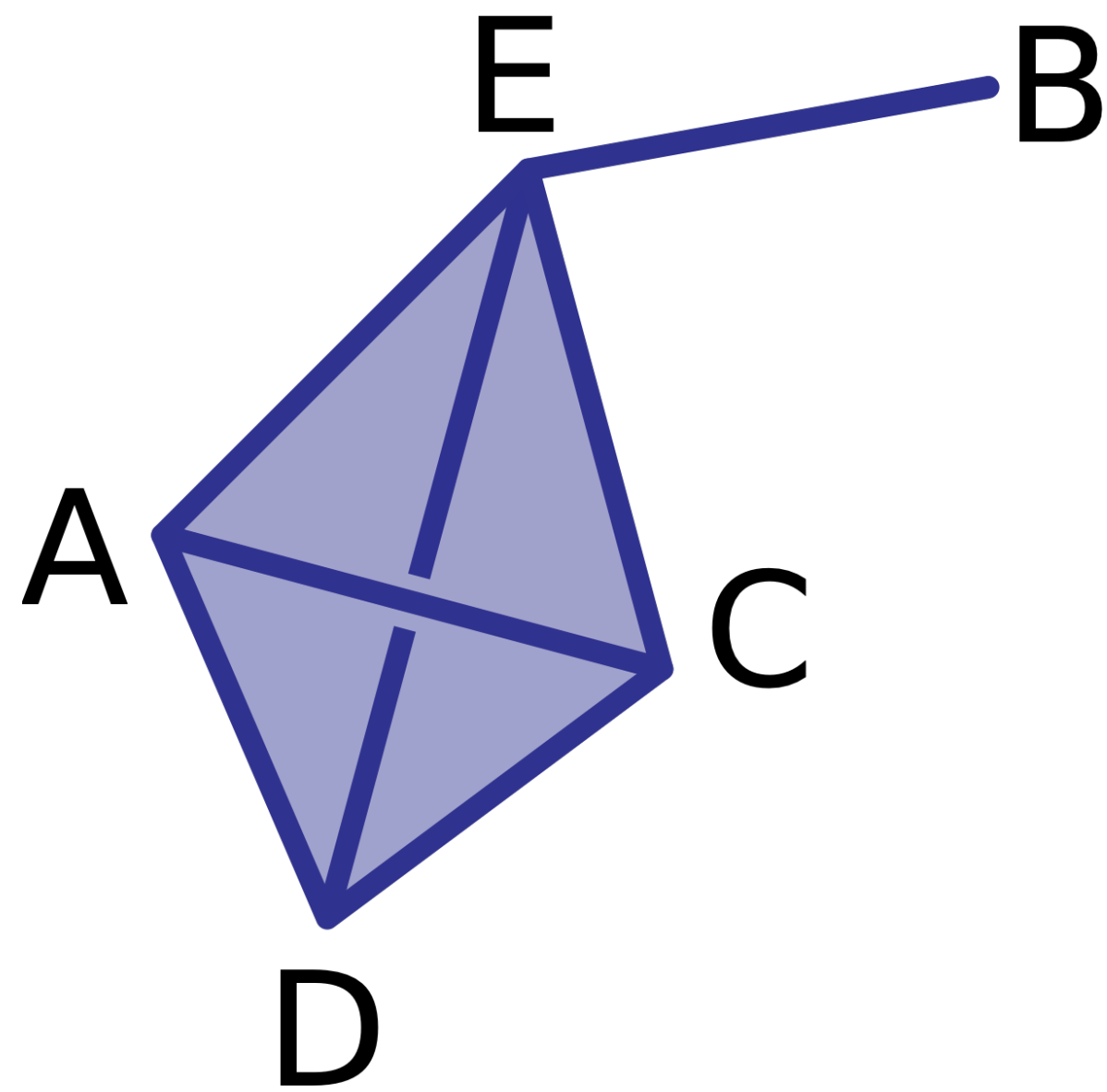
What are the generators of $B_1(K)$? Of $Z_1(K)$?

Example: Cycles and Boundaries



What are the generators of $B_1(K)$? Of $Z_1(K)$?

Example: Cycles and Boundaries



What are the generators of $B_1(K)$? Of $Z_1(K)$?

Homology for Real Now

Quotient Space

Let V be a vector space over a field k . Let $W \subset V$ be a subspace.

Define $\sim$ on V by $x \sim y$ iff $x - y \in W$. The equivalence class of x is denoted

$$[x] = x + W = \{x + w : w \in W\}$$

The quotient space V/W is then defined as $\{[x] \mid x \in V\}$. This is also a vector space with:

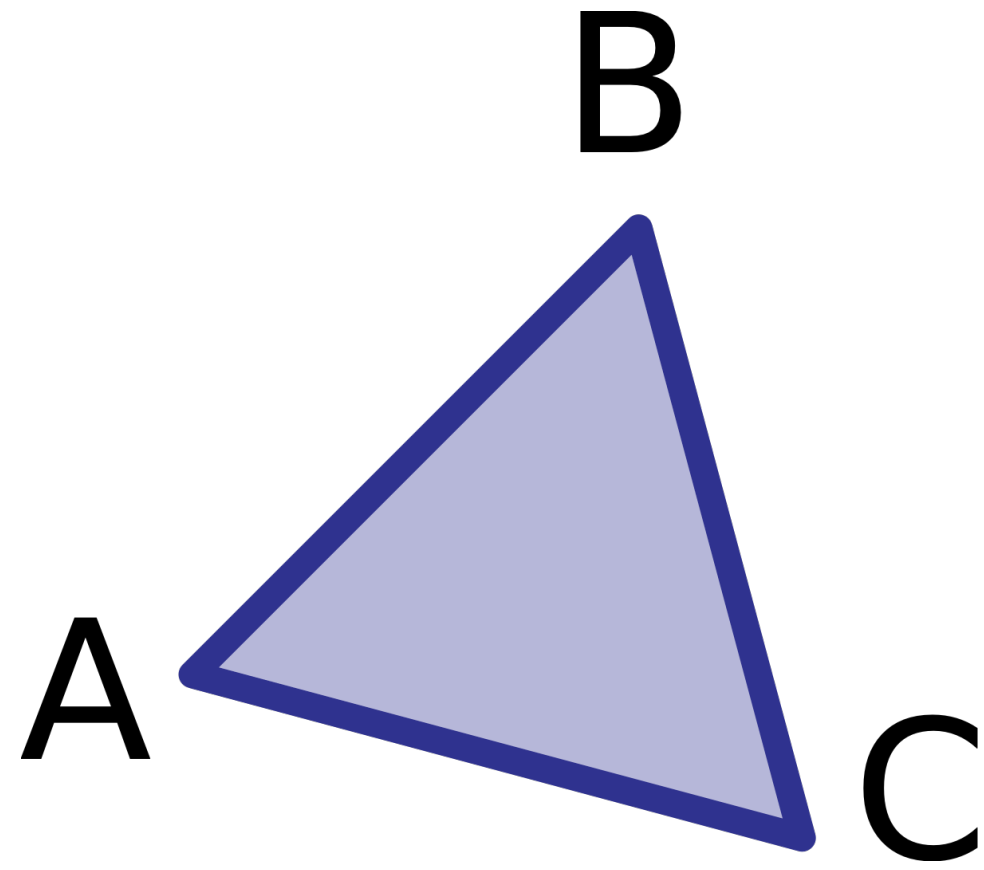
- Scalar multiplication:
- Addition:

Homology!!

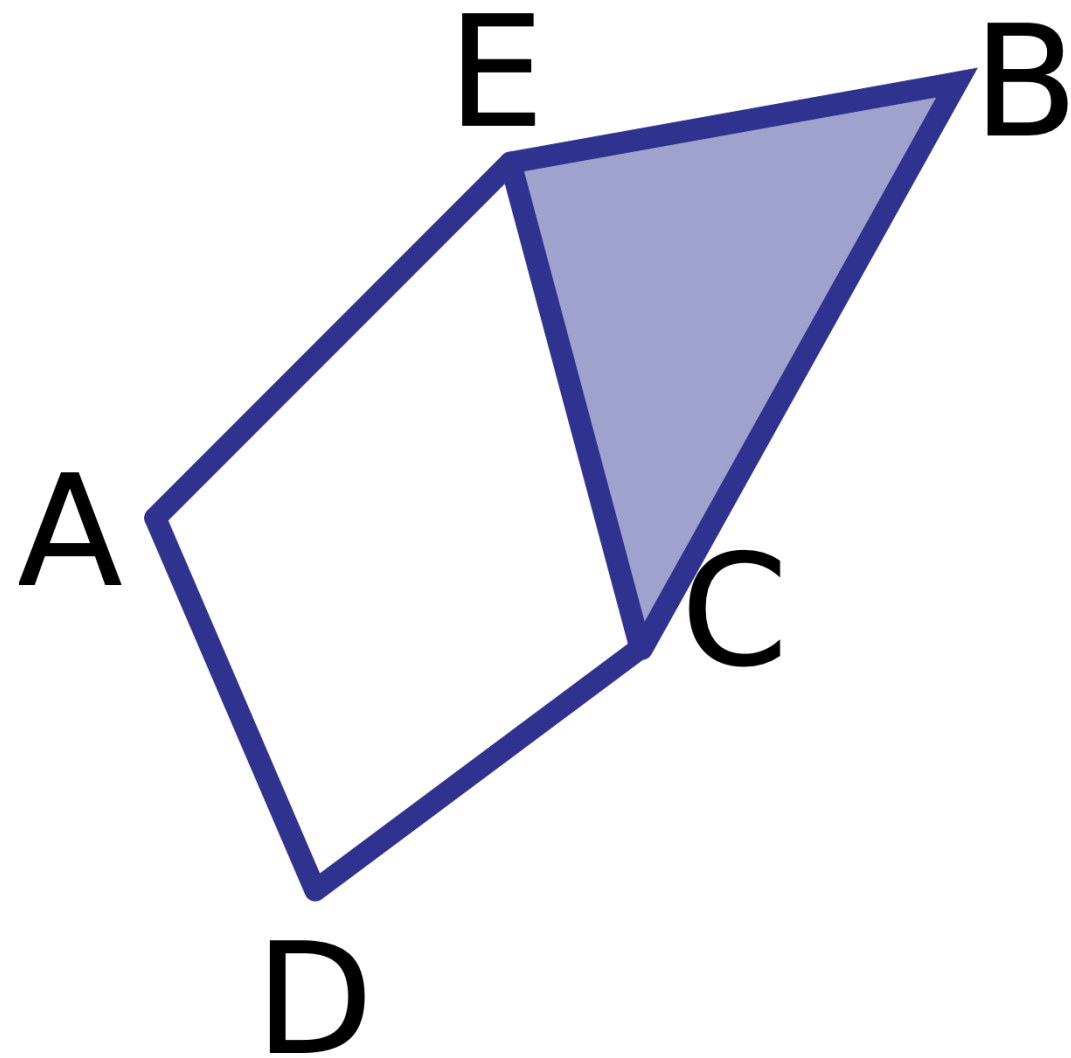
The *p*th homology group is the quotient space

$$H_p(K) := Z_p(K)/B_p(K)$$

Example: What is $H_1(K)$?



Example: What is $H_1(K)$?



Example: What is $H_2(K)$?

